

Question

Analytical Chemistry Calculation of the Feasibility of Oxalate Complex Synthesis

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Oxalate is a good ligand that can form complexes with transition metals. Since oxalate solutions are strongly alkaline, easily hydrolyzed metal ions such as Fe^{3+} and Al^{3+} need to be first converted to hydroxides, then dissolved with oxalic acid, and neutralized with base to obtain the corresponding oxalate complexes. This lengthy procedure takes a long time, the difficulty of filtering the hydroxide is high, the operational losses are large, and the yield is often not ideal.

An experimental scheme has been proposed to directly prepare a complex of oxalate with ferric iron, which is shown below.

Mix 20 mL $3 \text{ mol} \cdot \text{L}^{-1}$ potassium oxalate hot solution and 8 mL $1.5 \text{ mol} \cdot \text{L}^{-1}$ ferric chloride solution, cool to room temperature to obtain green needle-like crystals of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$. The volume of the solution at this time was measured to be 30 mL.

For this scheme, we can judge its rationality and feasibility by calculating the relative amounts of the main product and by-products (mainly $\text{Fe}(\text{OH})_3$).

Constants involved in this question:

$$\text{p}K_{\text{sp}}(\text{Fe}(\text{OH})_3) = 38.55$$

$$\lg K_{\text{e}}(\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}) = 20.2$$

$$\text{p}K_{\text{bl}}(\text{C}_2\text{O}_4^{2-}) = 9.73.$$

The saturated solubility of the product at room temperature is $4.9 \text{ g} \cdot 100 \text{ mL}^{-1}$, and the solubility of the two reactants is very high.

Let's first assume that $\text{Fe}(\text{OH})_3$ and other iron-containing species are not formed. After performing this synthesis operation (without separating the solid product from the solution), according to the assumption

above, we can calculate the concentrations of Fe^{3+} , $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$, $\text{C}_2\text{O}_4^{2-}$, HC_2O_4^- , and OH^- in the solution when it is cooled to room temperature (ignoring the secondary hydrolysis of oxalate).

Based on the preliminary calculation of the above species concentrations, we can judge whether $\text{Fe}(\text{OH})_3$ will be formed. So, please determine whether $\text{Fe}(\text{OH})_3$ will be formed. If so, calculate the mass of $\text{Fe}(\text{OH})_3$ formed (choose the answer closest to the calculated value).

- A.** Completely no $\text{Fe}(\text{OH})_3$ is generated.
- B.** $\text{Fe}(\text{OH})_3$ will be generated, with a mass of 0.13 g.
- C.** will generate $\text{Fe}(\text{OH})_3$, with a mass of 0.11 mg
- D.** will produce $\text{Fe}(\text{OH})_3$, with a mass of 3.7×10^{-22} g
- E.** will generate $\text{Fe}(\text{OH})_3$, with a mass of 5.7×10^{-22} g
- F.** will generate $\text{Fe}(\text{OH})_3$ with a mass of 0.36 mg.
- G.** will generate $\text{Fe}(\text{OH})_3$, with a mass of 0.18 mg

Answer

Correct Answer: **C**

Detailed Explanation

Due to the large stability constant of ferrioxalate, it can be assumed that most of the iron ions exist in the form of complexes.

First, assume that iron hydroxide is not generated.

Calculate the concentration based on the volume ratio: $[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}]_0 = 0.40 \text{ mol} \cdot \text{L}^{-1}$

Subtract the part of oxalate consumed to form the complex: $[\text{C}_2\text{O}_4^{2-}] = 0.8 \text{ mol} \cdot \text{L}^{-1}$

CHECKPOINT

1 PTS

$$[\text{C}_2\text{O}_4^{2-}] = 0.8 \text{ mol} \cdot \text{L}^{-1}$$

Considering the solubility of the product in water, calculate the saturation concentration:

$$[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}] = 0.1 \text{ mol} \cdot \text{L}^{-1}$$

CHECKPOINT

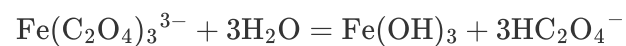
1 PTS

$$[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}] = 0.1 \text{ mol} \cdot \text{L}^{-1}$$

Calculate the concentration of free iron ions based on the stability constant: $[\text{Fe}^{3+}] \equiv 10^{-20.9} \text{ mol} \cdot \text{L}^{-1}$

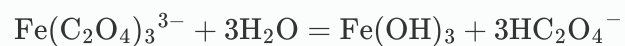
Based on the first-level hydrolysis of oxalate, we get: $[\text{HC}_2\text{O}_4^-] = [\text{OH}^-] = 10^{-4.91} \text{ mol} \cdot \text{L}^{-1}$

Next, consider the reaction of hydrolysis to form iron hydroxide:



CHECKPOINT

1 PTS



$$K = K_b^3 \cdot K_{\text{sp}} \cdot K_E = 10^{-10.84}, Q = 10^{-13.7}$$

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1 PTS

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$Q < K$, $\text{Fe}(\text{OH})_3$ will be generated.

CHECKPOINT

1 PTS

$Q < K$, $\text{Fe}(\text{OH})_3$ will be generated.

At equilibrium, $[\text{HC}_2\text{O}_4^-]_{\text{eq}} = 10^{-3.95} \text{ mol} \cdot \text{L}^{-1}$

Amount of precipitated substance:

$$n(\text{Fe}(\text{OH})_3) = ([\text{HC}_2\text{O}_4^-]_{\text{eq}} - [\text{HC}_2\text{O}_4^-]) \times 0.030/3 = 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

CHECKPOINT

2 PTS

$$n(\text{Fe}(\text{OH})_3) = ([\text{HC}_2\text{O}_4^-]_{\text{eq}} - [\text{HC}_2\text{O}_4^-]) \times 0.030/3 = 10^{-6} \text{ mol} \cdot \text{L}^{-1}$$

That is, the mass of the precipitate is $m = 0.11 \text{ mg}$